

Regional climate risk assessment from climate models using probabilistic machine learning

Received: 12 May 2026

Accepted: 26 August 2026

Published online: 28 September 2026

 Check for updates

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Effective climate risk assessment is hindered by the resolution gap between coarse global climate models and the fine-scale information needed for regional decision making. Existing downscaling methods struggle to bridge this gap: physics-based methods are too computationally expensive for large ensemble sampling of extreme events, while traditional statistical methods fail to capture multivariate spatiotemporal dependencies crucial to compound risk estimation. Furthermore, current machine learning approaches rely on temporally aligned training pairs, which are unavailable for free-running climate projections. Here we introduce GenFocal, an AI framework that generates statistically accurate, fine-scale weather from coarse climate projections without requiring paired simulated and observed events during training. GenFocal synthesizes complex and long-lived hazards, such as heatwaves and tropical cyclones, even when they are not well represented in the coarse climate projections. It also samples high-impact, rare events more accurately than leading methods. By translating large-scale climate projections into actionable localized information, GenFocal provides a powerful paradigm to improve climate adaptation and resilience strategies.

Effective economic and societal planning over horizons from years to decades is increasingly hinging on understanding and predicting how climate changes. While global climate models (GCMs) provide robust projections of large-scale trends, their coarse resolution creates a critical information gap for local and regional decision-making¹. This gap limits risk assessments for infrastructure design², energy system planning³, flood forecasting⁴ and financial services such as insurance pricing⁵. The challenge is particularly acute when assessing compound events, such as concurrent heat and drought, where complex spatiotemporal correlations are not resolved by GCMs, yet drive the most severe effects^{6–8}. Bridging this scale gap is a grand computational

and scientific challenge, and the primary obstacle to translating climate science into actionable intelligence for climate adaptation and resilience strategies.

This global-to-regional translation is fundamentally a problem of characterizing the statistics of fine-scale physical processes from coarse-grained climate projections. The large computational cost of GCMs restricts them to coarse grids, which inherently limits their fidelity and introduces biases. Furthermore, the chaotic nature of the climate system means that directly aligning coarse climate simulations with local weather observations at fine temporal scales is impossible. Historically, this lack of alignment has seriously hampered the

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development of a general statistical mapping from GCM outputs to consistent local weather that reliably preserves the complex correlation structures needed for accurate regional climate risk assessment. In particular, this consistency with local weather is essential to near-term climate risk assessment where the statistical characterization must be grounded in past observations⁸.

Existing efforts to overcome this downscaling challenge aim to correct biases and add coherent, fine-scale detail to coarse GCM outputs. Physics-based dynamical downscaling uses regional climate models forced by GCMs to simulate climate processes at higher resolution. They capture the rich spatiotemporal correlations of climate processes, but their computational expense limits the uncertainty quantification and robust assessment of extreme events, which requires a large number of samples^{9,10}. Statistical downscaling methods are a computationally efficient alternative, but they often lack the flexibility to capture the full range of spatiotemporal correlations that define complex weather and compound events¹¹. Recent advances in machine learning have spurred a new generation of machine learning-based downscaling methods that have shown great promise in weather forecasting and regional climate model emulation^{12–14}. However, these methods predominantly rely on a supervised learning paradigm that requires temporally aligned pairs of low-resolution and high-resolution data for training. This requirement poses a substantial challenge for downscaling climate projections, for which corresponding high-resolution ground truth data are unavailable.

We propose GenFocal, a computationally efficient, general-purpose generative artificial intelligence (AI) framework that addresses this grand challenge. GenFocal rapidly generates large ensembles of weather realizations that are physically realistic and statistically consistent with conditioning coarse climate projections. It is capable of elucidating complex, fine-scale weather phenomena from coarse climate data sources where those phenomena are poorly resolved.

GenFocal yields higher-fidelity local climate information than well-established statistical methods, including those used in leading reports such as the US Fifth National Climate Assessment. By modelling spatial and temporal correlations, GenFocal can capture the risk and evolution of spatiotemporal extremes—such as heatwaves and tropical cyclones (TCs)—in a unified framework: a capability that is well beyond current statistical models.

GenFocal

Figure 1 provides a schematic of the algorithmic pipeline for GenFocal, highlighting its key innovations (see Supplementary Information, section 9 for complete details). GenFocal incorporates three main features, namely two distinct generative AI techniques for bias correction and super-resolution, alongside the explicit modelling of temporal sequences of consecutive weather states at both the coarse-scale and fine-scale levels. Together, these features make GenFocal capable of probabilistically mapping coarse climate projections to fine-grained weather fields with strong temporal coherence, enabling the accurate characterization of complex, multi-day events such as prolonged heatwaves or TCs.

In this work, GenFocal downscales coarse climate states (y in Fig. 1) from 1.5° resolution to fine-grained weather states (x in Fig. 1) at 0.25° resolution in a localized region, using an intermediate latent space (y' in Fig. 1) of coarse-grained weather states matching the resolution of y . The input features 10 daily averaged variables, while the output contains 4 variables sampled 2-hourly (Supplementary Table 5). GenFocal is trained on 20 years (1980–1999) of data, using the publicly available ERA5 reanalysis¹⁵ as the high-resolution target and the Community Earth System Model Version 2 (CESM2) Large Ensemble (LENS2)¹⁶ as the coarse-resolution source. Hyperparameter tuning was performed using the period 2000–2009, and the final evaluation is reported for the 10-year period 2010–2019, downscaling the full LENS2 ensemble.

This chronological split for training, validation and testing aligns with the intended application of assessing climate risk over the next few decades⁸. Details can be found in Supplementary Information, section 6 (including the data used) and Supplementary Information, section 9 (including details on the frameworks, neural architectures, training and hyperparameter tuning).

Catalysed by the recent emergence of generative AI models, a few recent studies have explored generative modelling in weather forecasting^{17,18} and downscaling¹². We provide a detailed review of these modern machine learning techniques, alongside traditional methods, in Supplementary Information, section 1. A key limitation of existing approaches is their reliance on temporally aligned source and target data, generally unavailable when bridging coarse climate simulations with historical weather records. GenFocal is designed to overcome the scientific challenge of correlational modelling of chaotic systems by operating without this alignment assumption. It offers the scalability needed for real-world datasets and has undergone thorough methodological verification.

Realistic genesis and evolution of TCs

TCs are exceptionally destructive natural hazards responsible for thousands of deaths and tens of billions of dollars in damages every year. The success of mitigation strategies depends heavily on reliable projections of TC frequency, intensity and tracks under different climate scenarios. High-fidelity simulation of fine-grained physical processes is necessary for driving TC genesis and evolution, requiring much higher resolutions than those afforded by current GCMs. Physics-based dynamical downscaling via regional climate models can accurately capture the evolution of individual TCs, but it remains too expensive to generate the vast amount of data necessary to assess regional TC risk¹⁹. Studying future TC risk with statistical downscaling is possible, but only through bespoke methods that do not capture their interaction with the environment, and emulate TCs with reduced order systems that are partially coherent with their underlying physics²⁰.

By contrast, GenFocal is able to capture the full life cycle of TCs, from genesis to maturity (Supplementary Fig. 2), without specifically targeting these emergent extreme phenomena in our model design and training. As shown in Fig. 2 for the North Atlantic basin, GenFocal is able to generate TCs on the basis of the input's large-scale conditions, even when these storms are largely absent from the input climate projections (Methods and Supplementary Information, section 8). This ability crucially broadens GenFocal's applicability compared to methods reliant on input data at resolutions beyond those routinely available from climate models^{13,20}.

GenFocal generates TCs with tracks (Fig. 2d,g), cyclogenesis locations (Fig. 2b), landfalls (Fig. 2c), frequency (Fig. 2h), intensity (Fig. 2f,i) and morphology (Fig. 2e and Supplementary Fig. 9) consistent with the ERA5 reanalysis and the target resolution²¹ over the test period 2010–2019. This is in stark contrast with the statistics and tracks identified in the coarse LENS2, which exhibit both a lower frequency and excessively long durations (Fig. 2a,e).

Accurate assessment of compound climate risk

The risk of compound extremes arises from the cumulative effect of interacting physical processes, such as wildfires fuelled by dry vegetation and fanned by strong winds. This type of interdependency is often underestimated by downscaling methods that neglect correlations between hazards and their timescales^{6,22}. Humid heatwaves, characterized by prolonged periods of high temperature and humidity, are among the most frequent and impactful of such events, straining human health and power grids. We evaluate the ability of GenFocal to represent humid heatwaves by analysing the risk of summer heat index extremes in the Conterminous United States (CONUS) across timescales (Fig. 3). The physical and spatial structure of heatwaves is further examined in terms of the tail dependence of temperature

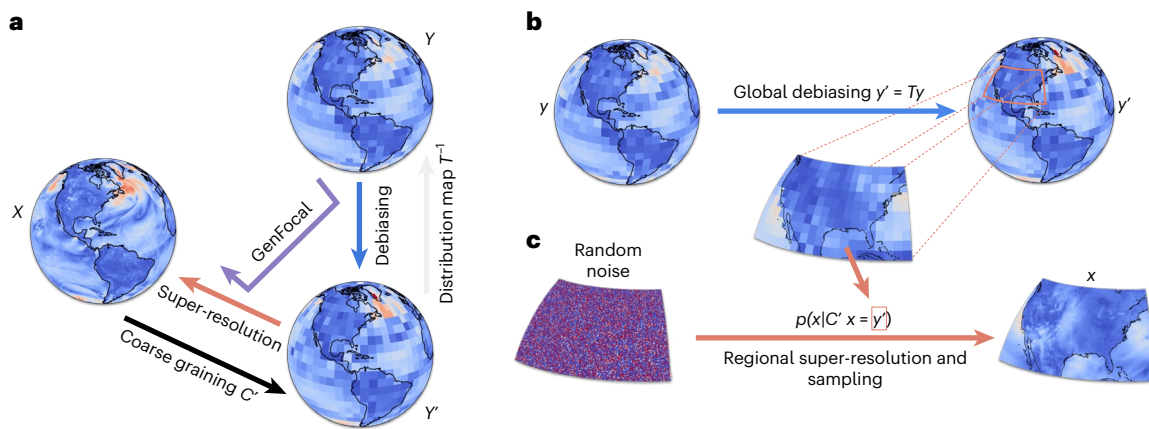


Fig. 1 | GenFocal is an end-to-end generative AI framework for climate downscaling, transforming coarse climate projections into actionable, fine-scale information for local regions, thereby enabling climate risk assessment and adaptation. **a**, Two-stage process. First, a coarse climate simulation from the space Y is bias-corrected into the low-resolution space y' in the same spatiotemporal grid (daily mean at 1.5°). A super-resolution step then increases the resolution from y' to the target weather-state space X . **b**, The debiasing operator T is instantiated as a rectified flow⁴⁸ to match the

distributions of the global low-resolution climate and a latent low-resolution weather space. **c**, The super-resolution step, $p(x|y')$, uses a conditional diffusion model⁴³ to statistically invert the coarse-graining map C' in a localized region, for example CONUS or North Atlantic basin. This process adds fine-grained spatiotemporal details, increasing temporal resolution from daily to 2-hourly within a local patch. At inference, a domain decomposition technique ensures temporal consistency across long sequences (Supplementary Information, section 9.5.3 and Supplementary Fig. 30).

and humidity extremes and the spatial autocorrelation, respectively. (For the definitions of these metrics, see Supplementary Information, section 7.)

GenFocal yields accurate estimates of the 99th percentile of the heat index during the summer months, with an average bias reduction over 35% with respect to the statistical downscaling baselines, which systematically underestimate risk (Fig. 3f,k,p). Furthermore, the tail dependence of temperature and humidity extremes demonstrates its superior ability to capture concurrent hazards, with notable improvements across the midwest and the western USA (Fig. 3g,l,q). These improvements amount to an average error reduction of 44% with respect to Seasonal Trends and Analysis of Residuals Empirical-Statistical Downscaling Model (STAR-ESDM) and Bias Correction and Spatial Disaggregation (BCSD). GenFocal also reproduces the spatial structure of weather patterns, which is strongly affected by fine-scale processes characteristic of regions with diverse topography such as California. The spatial correlations of the heat index and wind speed over this region with respect to San Francisco are shown in Fig. 3d,e, evaluated from the ERA5 reanalysis data. GenFocal captures the summertime decorrelation in the heat index between San Francisco and inland California driven by the coastal cooling effect of sea breeze, which increases with inland temperatures²³ (Fig. 3i). GenFocal also reproduces the complex spatial correlations of wind speed modulated by changes in topography (Fig. 3j). Downscaling methods that do not model spatial correlations explicitly, such as BCSD and STAR-ESDM, typically fail to identify the rich spatial correlation structure from the coarse climate simulation (Fig. 3n,o,s,t).

Heat-related mortality increases with heatwave duration²⁴, highlighting the importance of estimating the risk of extended periods of extreme heat. Capturing persistent events requires adequate representation of the temporal coherence of climate fields, which GenFocal models explicitly. We assess the skill at predicting extended heatwaves by estimating the risk of 5-day streaks with daily maximum heat indices exceeding 305 K. This threshold corresponds to the 'extreme caution' heat advisory of the National Oceanic and Atmospheric Administration. GenFocal provides largely unbiased estimates of 5-day extreme caution heat streaks across the east coast and the midwest, compared with the statistical downscaling methods, which tend to overestimate risk in these regions (Fig. 3h,m,r). Overall, GenFocal reduces average bias by 44% and 57% compared to BCSD and STAR-ESDM, respectively.

The skilful estimation of compound climate risks by GenFocal, demonstrated here for heatwaves and previously for TCs, stems in great measure from its ability to capture correlations across meteorological fields, space and time. In addition, the risk estimates provided by GenFocal benefit from a more accurate representation of the marginal distribution of directly modelled fields than other methods. For example, GenFocal reduces the bias of the 99th percentile of near-surface temperature and humidity by more than 20% and 22%, respectively (Supplementary Fig. 13). Further results are presented in Supplementary Information, section 3.

Future climate risk assessment

The design of critical infrastructure with expected lifetimes of decades to centuries requires an assessment of future climate risk. To provide reliable assessments, downscaling methods must not only preserve trends projected by the input coarse climate data but also capture the effects of those changes to weather phenomena unresolved by the original projections. Preserving climate change signals can be challenging for statistical downscaling methods trained to correct biases over a reference historical period, due to the distortion of climate change trends in the debiasing process¹¹.

We assess the ability of GenFocal to evaluate future climate risk by analysing projected changes in summer heat extremes across cities in the western USA, and trends in TC activity in the North Atlantic basin.

Changes in summer heat extremes in the western USA

The western USA is expected to experience a substantial increase in extreme heat severity in the coming decades²⁵. We evaluate the climate change response of summer temperature extremes projected by GenFocal by comparing them to dynamically downscaled climate projections from the Western United States Dynamically Downscaled Dataset²⁶. Although dynamical downscaling is also subject to model errors, its reliance on physics-based modelling relaxes stationarity assumptions and ensures physically consistent climate change patterns²⁷. The dynamical downscaling simulations considered use the Weather Research and Forecasting model and take as input data from the same climate model, CESM2, debiased a priori using the ERA5 reanalysis. We report results for dynamically downscaled projections at 45-km and 9-km resolutions, after interpolation to the resolution of GenFocal, to illustrate variability due to fine-scale processes.

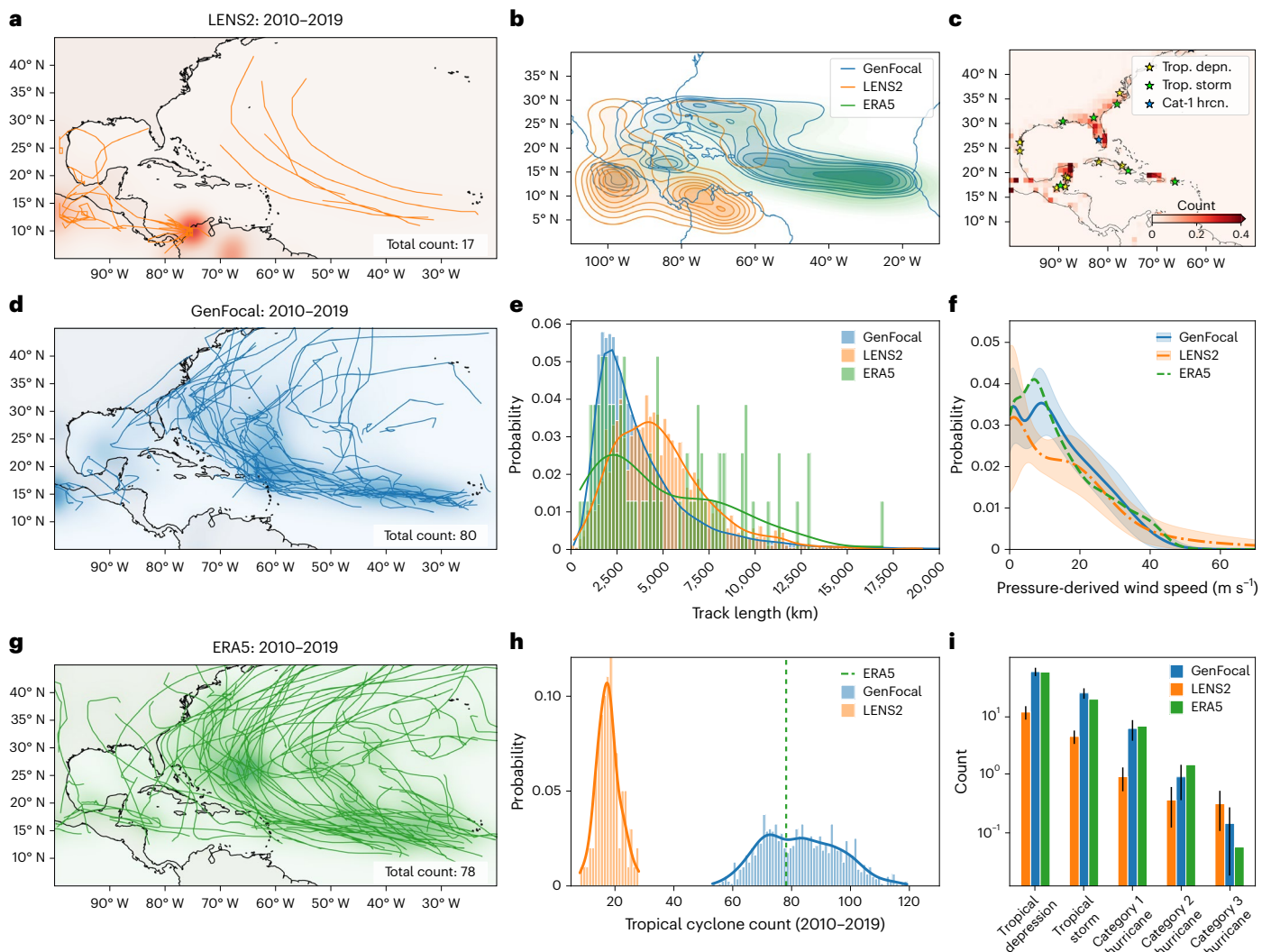


Fig. 2 | GenFocal accurately reproduces the statistics of TCs in the North Atlantic in the time period 2010–2019, in terms of cyclogenesis, intensity and morphology. **a**, Ensemble track density and tracks for a single member from LENS2. **d**, Ensemble track density and tracks for a single downscaled high-resolution member generated by GenFocal. **g**, Tracks and density map from the historical ERA5 reanalysis. **b**, Kernel density estimates of cyclogenesis locations. **e**, Length of the tracks and their kernel density estimate histograms,

characterizing their morphology. **c**, Expected landfall count overlaid with ERA5 observations. Cat, category; depn., depression; hrcn., hurricane; trop., tropical. **f**, Distribution of pressure-derived wind speed with two ensemble standard deviation confidence intervals. **h**, TC count and their kernel density estimate histograms. **i**, Saffir–Simpson scale distributions with error bars denoting one ensemble standard deviation. For LENS2 and GenFocal, we use 100 and 800 members, respectively, to compute ensemble standard deviations.

Figure 4 evaluates changes in the top decile of daily maximum temperature across different cities of the western USA over the period 2020–2080, a complex statistic that requires spatiotemporal downscaling of the input daily averaged climate data. Results for additional cities and statistics are included in Supplementary Information, section 4. GenFocal exhibits similar regional warming trends to Weather Research and Forecasting, with relatively weak warming in coastal San Diego and much stronger warming trends in inland cities such as Albuquerque, Phoenix and Portland. BCS and STAR-ESDM fail to capture this modulation of climate change by regional processes, predicting quasi-uniform warming across regions.

Projecting future TC risk

GenFocal demonstrates the ability to realize detailed TC activity driven by climate change, based on the underlying large-scale conditions, even when these specific events are not explicitly resolved by the input coarse climate simulations. To show this, we evaluate trends from 2010–2019 to 2050–2059 by producing downscaled results covering 8,000 August–October seasons representative of each period with

GenFocal: we downscale 10-year trajectories from the LENS2 ensemble with 8 samples per trajectory.

Over the first half of the 21st century, GenFocal projects an increase in the number of tropical storms and hurricanes making landfall over the US east coast (Fig. 5a,d). This projection aligns with forecasts from other downscaled climate projections, such as the Risk Analysis Framework for Tropical Cyclones model²⁸. These findings contribute to the ongoing scientific investigation and refinement of understanding regarding North Atlantic TC landfall trends. GenFocal also predicts subtropical intensification and tropical weakening of TCs over the North Atlantic basin (Fig. 5e,f), consistent with the observed poleward migration of the location of TC maximum intensity^{29,30}. The projected TC intensification is largest over the Carolinas and the Mid-Atlantic, with the most intense TCs projected to strengthen at a faster pace (Supplementary Fig. 29).

Discussion

GenFocal leverages generative AI to overcome the limitations of traditional climate downscaling methods. It is trained directly on coarse climate simulations and weather reanalysis data, eliminating the need

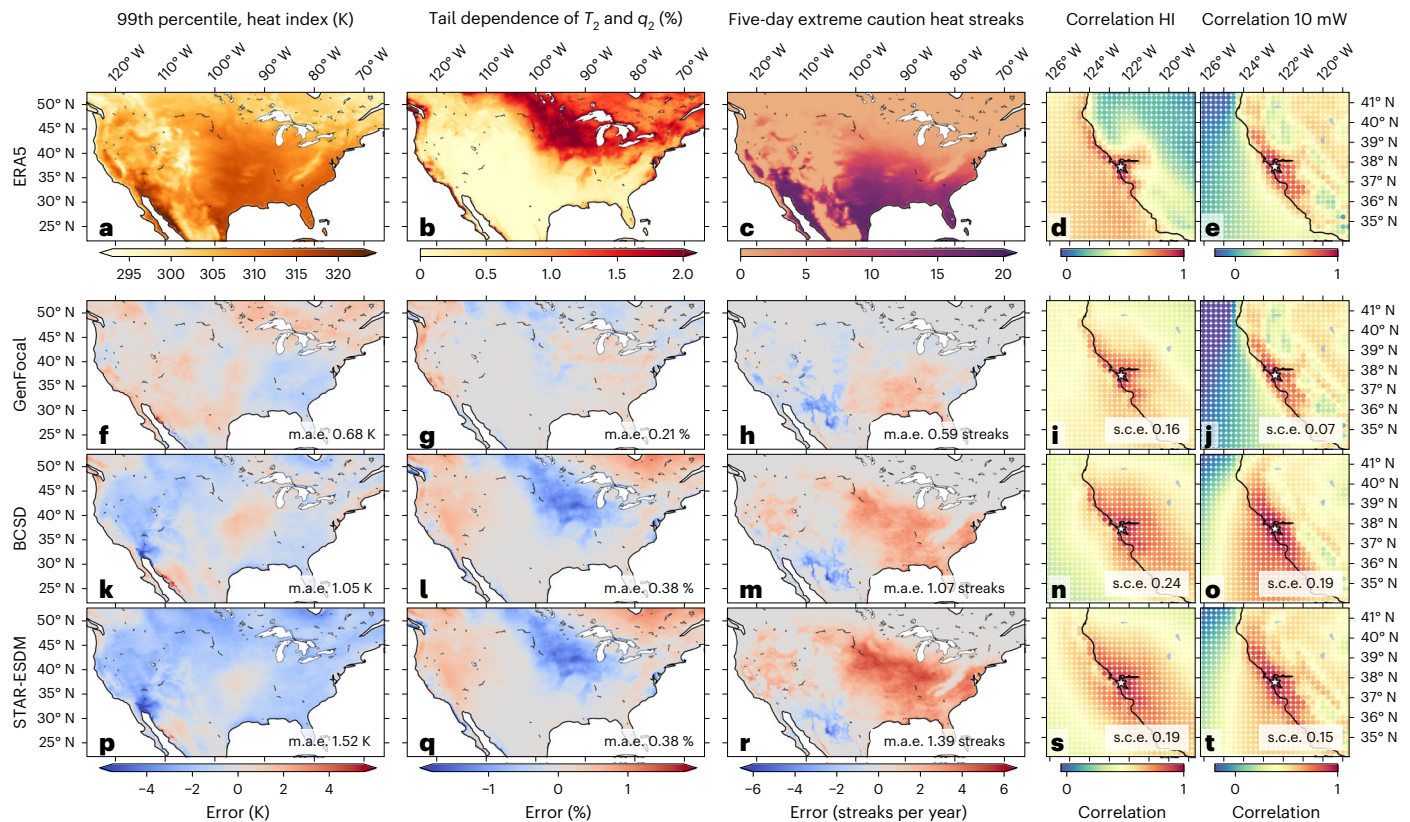


Fig. 3 | GenFocal accurately assesses projected compound heat extremes over the CONUS during the summer (June–August) of the evaluation period 2010–2019. GenFocal outperforms competing approaches, the BCSD^{49,50}, a method routinely used for downscaling ensembles from the Coupled Model Intercomparison Project (CMIP)³¹ and the STAR-ESDM⁵², a state-of-the-art method recommended for use in the US Fifth National Climate Assessment⁵³. For details about those two methods, see Supplementary Information, section 5. **a**, Heat index 99th percentile. **b**, Tail dependence of 2-metre temperature

(T_2) and specific humidity (q_2) extremes. **c**, Number of 5-day streaks with ‘extreme caution’ heat advisory per year. **f–h, k–m, p–r**, Errors in downscaled estimates are shown for GenFocal (**f–h**), BCSD (**k–m**) and STAR-ESDM (**p–r**). **d, i, n, s**, Spatial correlation of the heat index (HI) of San Francisco and its surroundings, evaluated at 18Z for ERA5 (**d**), GenFocal (**i**), BCSD (**n**) and STAR-ESDM (**s**). **e, j, o, t**, Spatial correlation of the 10-m wind speed for ERA5 (**e**), GenFocal (**j**), BCSD (**o**) and STAR-ESDM (**t**). Insets show the mean absolute error (m.a.e.) and spatial correlation error (s.c.e.) of the downscaled results.

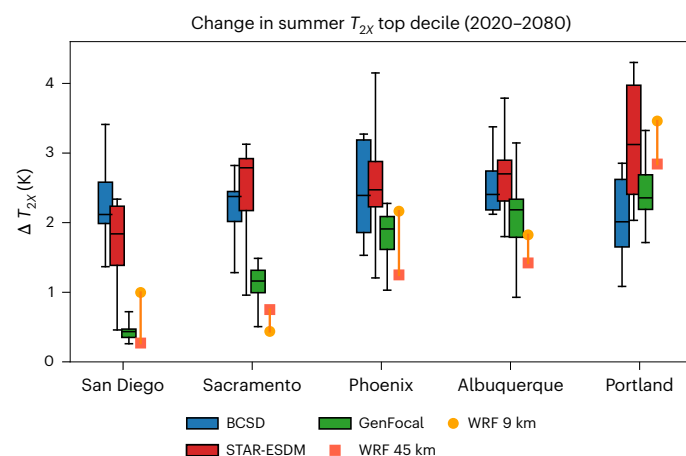


Fig. 4 | GenFocal projects future regional warming trends from coarse climate simulations, capturing the change patterns in extreme heat across cities in the western USA (2020–2080) more consistently than other methods. Shown is the top decile of daily maximum near-surface temperature. Results are computed as the average over $1^\circ \times 1^\circ$ regions, and 7 summers (June–August) centred around 2020 and 2080. Boxes for BCSD, STAR-ESDM and GenFocal show the median (50th percentile) and interquartile range (25th and 75th percentiles) of an ensemble of eight projections, and whiskers represent the ensemble minima and maxima.

for costly regional climate model simulations to establish temporally aligned training data²² while explicitly accounting for coarse model biases. By leveraging historical reanalysis to correct these systematic biases, GenFocal provides a means to observationally constrain future climate projections, analogous to recent transfer learning frameworks that blend historical observations with multi-model ensembles to reduce projection uncertainty³¹. Moreover, it is designed to capture the spatiotemporal, multivariate statistics of climate data accurately, addressing key limitations of statistical downscaling methods, such as BCSD and STAR-ESDM, which are incapable of modelling the complex interdependency of multiple variables. This enables GenFocal to quantify the uncertainty of TCs and other compound extreme events robustly and efficiently. Such tasks have traditionally required narrowly focused, bespoke statistical emulators or computationally expensive dynamical downscaling.

The practical implications of our method are significant for downstream applications that demand physically consistent localized climate data. For instance, accurate spatial correlation modelling can improve system-wide energy grid planning by estimating the risk of concurrent heat extremes that increase energy demand and vulnerability of power lines^{3,32}. In addition, the ability to capture inter-variable correlations, such as those between temperature and humidity, is essential for predicting the heat index, which has direct applications in public health, food production³³, energy demand forecasting^{32,34} and disaster preparedness³⁵. Furthermore, directly modelling temporal

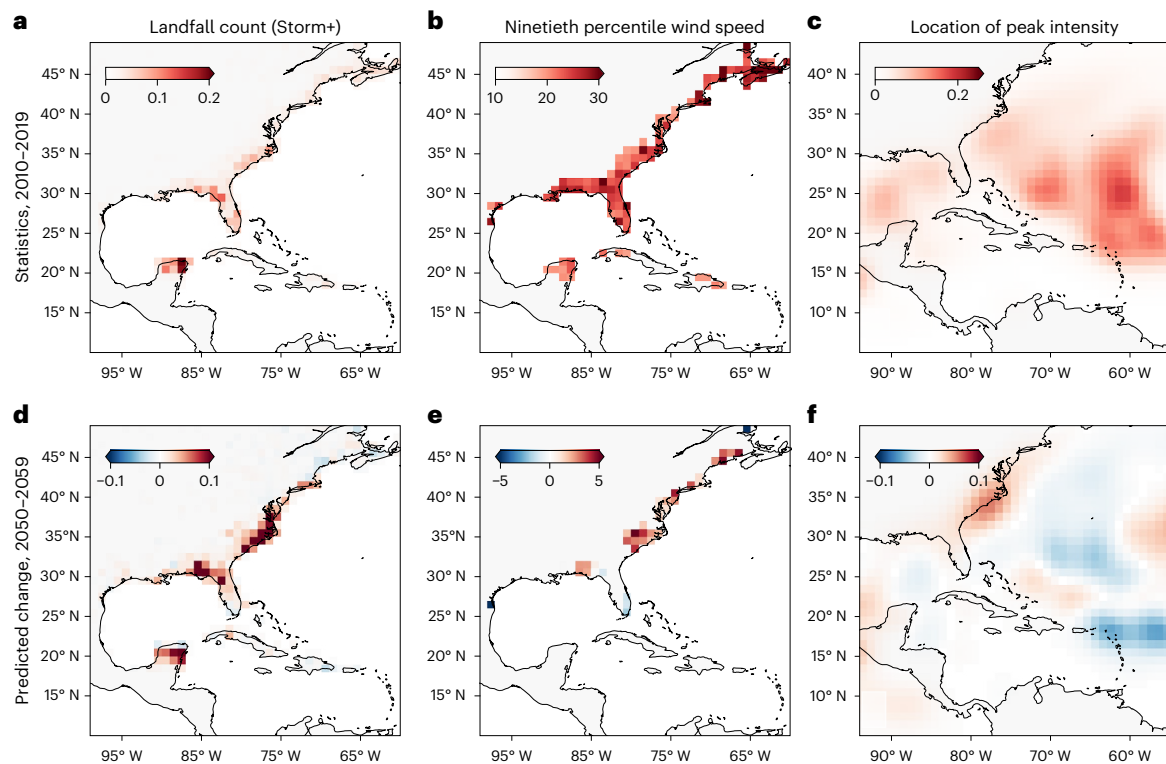


Fig. 5 | GenFocal projects trends in TC landfall frequency and intensity over the first half of the 21st century consistent with well-established methods²⁸ and recent trends^{29,30}. **a, d**, Number of tropical storm and hurricane landfalls during the August–October season of years 2010–2019 (**a**) and its projected change by 2050–2059 (**d**). **b, e**, Ninetieth percentile of maximum pressure-derived wind speed (m s^{-1}) of landfalling TCs (**b**) and its projected

change over the same periods (**e**). **c, f**, Spatial distribution of lifetime TC peak intensity (**c**) and its projected change over the same periods (**f**). All results are computed as the average over 800 downscaled climate projections, hence fractional counts. Changes are displayed only if they are statistically significant ($P < 0.05$ in a two-tailed Mann–Whitney U -test) and set to zero otherwise.

correlations improves risk estimates for extended extreme events, such as prolonged heatwaves and TCs, offering more reliable insights for resilience policies^{36,37}. By providing a full probabilistic characterization of climate effects, GenFocal enables assessing risks associated with compound hazards involving any number of meteorological extremes interacting across space and time.

Finally, GenFocal opens the way for downscaling efficiently large ensembles of climate projections, a computationally intractable task for physics-based downscaling approaches. This is a crucial capability for present and future risk assessments of regional extremes and rare events, such as TCs, particularly as advances in AI-accelerated climate simulation^{38–40} combined with advances in climate modelling and larger ensembles⁴¹ continue expanding our ability to sample global climate uncertainty.

Methods

Generative models used by GenFocal

GenFocal is a two-step framework: first, a temporal sequence of consecutive climate states, $y \in \mathcal{Y}$, which is coarse in scale and biased, is debiased into an intermediate sequence on the manifold y' that is consistent with a sequence of coarse-grained weather states Cx where $x \in \mathcal{X}$, the high-resolution weather manifold. A subsequent super-resolution step increases the spatiotemporal resolution of the debiased sequence while preserving temporal coherence. This two-stage design decouples learning the debiasing and the super-resolution operations, enabling ‘drop-in’ replacement of alternative debiasing operations, as explored in Supplementary Information, section 9.6.

Super-resolution. We construct C as a coarsening operation by down-sampling the ERA5 data from 2-hourly and 0.25° to daily and 1.5° , thus

forming pairs of aligned data samples ($y' = Cx_i, x_i$). To learn the super-resolution operation, that is, the inverse of the downsampling, we use a conditional diffusion model^{42,43}, popularized by latest advances in image and video generation. We take advantage of the previous knowledge that a spatially interpolated linear mapping $\mathcal{J}(y')$ already contains a strong approximation of the mean statistics of x by modelling the residual $r := x - \mathcal{J}(y')$. As such we use the conditional diffusion model to sample from $p(r|y')$ and then add the sampled residual back to $\mathcal{J}(y')$ to obtain the final output of the super-resolution.

The conditional diffusion model learns a neural network based denoiser to iteratively refine a noisy version of the residual $r + \epsilon\sigma$ to its clean version r . The noise is controlled by a scaled Gaussian variable $\epsilon \sim \mathcal{N}(0, 1)$ where the scale σ is sampled from a refinement scheduling distribution $\mathcal{Q}$. The denoiser D_θ is thus trained to minimize the loss function between the refined and the clean residuals:

$$\ell(\theta) = \mathbb{E}_{x \in \mu_x} \mathbb{E}_{\sigma \sim \mathcal{Q}(\sigma)} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, 1)} \|D_\theta(r + \epsilon\sigma, y') - r\|^2. \quad (1)$$

Once learned, the denoiser D_θ is used to construct a stochastic differential equation-based sampler that refines a Gaussian noise signal into a clean residual:

$$dr_\tau = -2\dot{\sigma}_\tau \sigma_\tau D_\theta(r_\tau, \sigma_\tau, y') d\tau + \sqrt{2\dot{\sigma}_\tau \sigma_\tau} d\omega_\tau, \quad (2)$$

in diffusion time τ from $\tau = \tau_{\max}$ to 0, and initial condition $r_{\tau_{\max}} \sim \mathcal{N}(0, \sigma_{\tau_{\max}}^2 I)$, where $\sigma_\tau : \mathbb{R} \rightarrow \mathbb{R}$ is the diffusion-time-dependent noise schedule, controlled by $\mathcal{Q}(\sigma)$. A more comprehensive description of the diffusion model is included in Supplementary Information, section 9.5.1, along with implementation details.

While the model D_θ is trained on short sequences such as one to a few days, we use an inference procedure to sample extended temporal sequences (spanning multiple months, for example). The procedure achieves temporal coherence through domain decomposition, where each shorter temporal period is a domain and overlapping domains are guided to coherence and contiguity. Details are provided in Supplementary Information, section 9.5.3.

Debiasing. Due to the lack of alignment between data sampled from y and y' , we seek a map between their sample distributions. This is a weaker notion than the sample-to-sample correspondence offered by physics-based downscaling methods. However, as demonstrated in this work, achieving a statistical distribution match can effectively debias while remaining computationally advantageous and generating plausible sampled states.

We leverage the idea of rectified flows⁴⁴ by constructing the debiasing map T as the solution map of an ordinary differential equation given by

$$\frac{dy}{d\tau} = \mathbf{v}_\phi(y, \tau) \quad \text{for } \tau \in [0, 1], \quad (3)$$

whose vector field $\mathbf{v}_\phi(x, \tau)$ is parametrized by a neural network (see Supplementary Information, section 9.4.3 for further details). By identifying the input of the map as the initial condition $y_0 = y(\tau=0)$, we have the solution as the mapping $T(y) := y(\tau=1)$. We train $\mathbf{v}_\phi$ by minimizing loss

$$\ell(\phi) = \mathbb{E}_{\tau \sim \mathcal{U}[0,1]} \mathbb{E}_{(y_0, y_1) \sim \pi \in \Pi(\mu_{y_0}, \mu_{y_1})} \| (y_1 - y_0) - \mathbf{v}_\phi(y_\tau, \tau) \|^2, \quad (4)$$

where $y_\tau = \tau y_1 + (1 - \tau) y_0$. $\Pi(\mu_{y_0}, \mu_{y_1})$ is the set of couplings observing the marginal distributions of y and y' respectively. Once $\mathbf{v}_\phi$ is learned, we debias any given y by solving (3) from $\tau=0$ to $\tau=1$ using the fourth-order Runge–Kutta ordinary differential equation solver.

Analogous to super-resolution, we also learn a debiasing map that takes into consideration a temporal sequence of climate variables. In Supplementary Information, section 9.4, we describe a simple way to achieve this as well as other important implementation details, such as selection of the coupling $\Pi(\mu_y, \mu_{y'})$ (see Supplementary Information, section 10.6 for an ablation study of different choices) and parametrization of $\mathbf{v}_\phi$ with various neural architecture choices.

Evaluation protocols and metrics

The downscaling methods are evaluated in two categories of metrics. The first set of metrics evaluates the discrepancy between the distributions of the downscaled climate data and the corresponding ERA5 weather data. Three types of discrepancy are measured. The first measures the univariate differences at each site, which are averaged in space to give rise to mean absolute bias, Wasserstein distance and percentile mean absolute error. The second measures spatial correlation and temporal spectrum errors. The last type measures correlation discrepancies among different variables such as tail dependence, an important quantity for compound extremes. Supplementary Information, section 7 provides detailed definitions of the evaluation metrics.

The second category of metrics is application-specific. In this work, we focus on North Atlantic TCs and severe and prolonged heat events over CONUS. In either case, non-trivial processing is performed on the output variables to compute composite variables (such as heat indices, the number of heat streak days) and TC occurrences and tracks. Evaluation metrics vary and we describe them in detail in Supplementary Information, section 8.

Data availability

Pretrained model weights are available via Google Cloud and Zenodo at <https://doi.org/10.5281/zenodo.21285802> (ref. 45). The data for training the models as well as debiased and downscaled forecasts produced

by GenFocal, are available via Google Cloud at <https://console.cloud.google.com/storage/browser/genfocal>. Dynamically downscaled projections from the WUS-D3 dataset are available via Amazon Web Services at <https://registry.opendata.aws/wrf-cmip6>.

Code availability

Source code for our models and evaluation protocols are available via Zenodo at <https://doi.org/10.5281/zenodo.21268077> (ref. 46) and GitHub at https://github.com/google-research/swirl-dynamics/tree/main/swirl_dynamics/projects/genfocal.

References

- Voosen, P. Local predictions of climate change are hazy. But cities need answers fast. *Science* **388**, 1016–1021 (2025).
- National Academies of Sciences, Engineering, and Medicine. *Modernizing Probable Maximum Precipitation Estimation* Technical report (NASEM, 2024).
- Qiu, L., Khorramfar, R., Amin, S., & Howland, M. F. Decarbonized energy system planning with high-resolution spatial representation of renewables lowers cost. *Cell Rep. Sustain.* **1**, 100263 <https://doi.org/10.1016/j.crsus.2024.100263> (2024).
- Nearing, G. et al. Global prediction of extreme floods in ungauged watersheds. *Nature* **627**, 559–563 (2024).
- Mills, E. Insurance in a climate of change. *Science* **309**, 1040–1044 <https://doi.org/10.1126/science.1112121> (2005).
- Zscheischler, J. et al. Future climate risk from compound events. *Nat. Clim. Change* **8**, 469–477 (2018).
- Bevacqua, E. et al. Advancing research on compound weather and climate events via large ensemble model simulations. *Nat. Comm.* **14**, 2145 (2023).
- Gettelman, A., Tebaldi, C. & Leung, L. R. Climate nowcasting. *Environ. Res. Clim.* **4**, 013002 (2025).
- Giorgi, F. Thirty years of regional climate modeling: where are we and where are we going next? *J. Geophys. Res. Atmos.* **124**, 5696–5723 (2019).
- Goldenson, N. et al. Use-inspired, process-oriented GCM selection: prioritizing models for regional dynamical downscaling. *Bull. Am. Meteorol. Soc.* **104**, E1619–E1629 (2023).
- Chandel, V. S., Bhatia, U., Ganguly, A. R. & Ghosh, S. State-of-the-art bias correction of climate models misrepresent climate science and misinform adaptation. *Environ. Res. Lett.* **19**, 094052 (2024).
- Mardani, M. et al. Residual corrective diffusion modeling for km-scale atmospheric downscaling. *Commun. Earth Environ.* **6**, 124 (2025).
- Lockwood, J. W., Gori, A. & Gentile, P. A generative super-resolution model for enhancing tropical cyclone wind field intensity and resolution. *J. Geophys. Res. Mach. Learn. Comput.* **1**, e2024JH000375 (2024).
- Rampal, N. et al. Enhancing regional climate downscaling through advances in machine learning. *Artif. Intell. Earth Syst.* **3**, 230066 (2024).
- Hersbach, H. et al. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* **146**, 1999–2049 (2020).
- Rodgers, K. B. et al. Ubiquity of human-induced changes in climate variability. *Earth Syst. Dyn.* **12**, 1393–1411 (2021).
- Price, I. et al. Probabilistic weather forecasting with machine learning. *Nature* **637**, 84–90 (2025).
- Li, L., Carver, R., Lopez-Gomez, I., Sha, F. & Anderson, J. Generative emulation of weather forecast ensembles with diffusion models. *Sci. Adv.* **10**, eadk4489 (2024).
- Jing, R., Lin, N., Emanuel, K., Vecchi, G. & Knutson, T. R. A comparison of tropical cyclone projections in a high-resolution global climate model and from downscaling by statistical and statistical-deterministic methods. *J. Clim.* **34**, 9349–9364 (2021).

20. Jing, R. et al. TC-GEN: Data-driven tropical cyclone downscaling using machine learning-based high-resolution weather model. *J. Adv. Model. Earth Syst.* **16**, e2023MS004203 (2024).
21. Davis, C. A. Resolving tropical cyclone intensity in models. *Geophys. Res. Lett.* **45**, 2082–2087 (2018).
22. Lopez-Gomez, I. et al. Dynamical-generative downscaling of climate model ensembles. *Proc. Natl Acad. Sci. USA* **122**, e2420288122 <https://doi.org/10.1073/pnas.2420288122> (2025).
23. Lebas, B. et al. Observed 1970–2005 cooling of summer daytime temperatures in coastal California. *J. Clim.* **22**, 3558–3573 (2009).
24. Brooke, A. G. & Bell, M. L. Heat waves in the United States: mortality risk during heat waves and effect modification by heat wave characteristics in 43 U.S. communities. *Environ. Health Perspect.* **119**, 210–218 <https://doi.org/10.1289/ehp.1002313> (2011).
25. Meehl, G. A. & Tebaldi, C. More intense, more frequent, and longer lasting heat waves in the 21st century. *Science* **305**, 994–997 (2004).
26. Rahimi, S. et al. An overview of the western United States dynamically downscaled dataset (WUS-D3). *Geosci. Model Dev.* **17**, 2265–2286 (2024).
27. Walton, D. et al. Understanding differences in California climate projections produced by dynamical and statistical downscaling. *J. Geophys. Res. Atmos.* **125**, e2020JD032812 (2020).
28. Balaguru, K. et al. Increased U.S. coastal hurricane risk under climate change. *Sci. Adv.* **9**, eadf0259 (2023).
29. Kossin, J. P., Emanuel, K. A. & Vecchi, G. A. The poleward migration of the location of tropical cyclone maximum intensity. *Nature* **509**, 349–352 (2014).
30. Studholme, J., Fedorov, A. V., Gulev, S. K., Emanuel, K. & Hodges, K. Poleward expansion of tropical cyclone latitudes in warming climates. *Nat. Geosci.* **15**, 14–28 (2022).
31. Immorlano, F. et al. Transferring climate change physical knowledge. *Proc. Natl Acad. Sci. USA* **122**, e2413503122 (2025).
32. Dumas, M., Kc, B. & Cunliff, C. I. *Extreme Weather and Climate Vulnerabilities of the Electric Grid: A Summary of Environmental Sensitivity Quantification Methods* Technical report (Oak Ridge National Laboratory, 2019).
33. Wang, B. et al. Sources of uncertainty for wheat yield projections under future climate are site-specific. *Nat. Food* **1**, 720–728 (2020).
34. Damiani, A., Ishizaki, N. N., Sasaki, H., Feron, S. & Cordero, R. R. Exploring super-resolution spatial downscaling of several meteorological variables and potential applications for photovoltaic power. *Sci. Rep.* **14**, 7254 (2024).
35. Goss, M. et al. Climate change is increasing the likelihood of extreme autumn wildfire conditions across California. *Environ. Res. Lett.* **15**, 094016 (2020).
36. Delworth, T. L., Mahlman, J. D. & Knutson, T. R. Changes in heat index associated with CO₂-induced global warming. *Clim. Change* **43**, 369–386 (1999).
37. Dahl, K., Licker, R., Abatzoglou, J. T. & Declet-Barreto, J. Increased frequency of and population exposure to extreme heat index days in the United States during the 21st century. *Environ. Res. Commun.* **1**, 075002 (2019).
38. Brenowitz, N. D. et al. Climate in a bottle: towards a generative foundation model for the kilometer-scale global atmosphere. Preprint at <https://doi.org/10.48550/arXiv.2505.06474> (2025).
39. Watt-Meyer, O. et al. ACE2: accurately learning subseasonal to decadal atmospheric variability and forced responses. *npj Clim. Atmos. Sci.* **8**, 205 (2025).
40. Henn, B. et al. AIMIP phase 1: systematic evaluations of AI weather and climate models. Preprint at <https://doi.org/10.48550/arXiv.2605.06944> (2026).
41. Dunne, J. P. et al. An evolving coupled model intercomparison project phase 7 (CMIP7) and fast track in support of future climate assessment. *Geosci. Model Dev.* **18**, 6671–6700 (2025).
42. Song, Y. & Ermon, S. Improved techniques for training score-based generative models. In *Proc. 34th International Conference on Neural Information Processing Systems* 12438–12448 (2020).
43. Song, Y. et al. Score-based generative modeling through stochastic differential equations. In *Proc. International Conference on Learning Representations* <https://openreview.net/pdf?id=PxTIG12RRHS> (2021).
44. Liu, Q. Rectified flow: a marginal preserving approach to optimal transport. Preprint at <https://doi.org/10.48550/arXiv.2209.14577> (2022).
45. Wan, Z. Y., Zepeda-Núñez, L. & Gomez, L. Model checkpoints for 'regional climate risk assessment from climate models using probabilistic machine learning'. *Zenodo* <https://doi.org/10.5281/zenodo.21285802> (2026).
46. Wan, Z. Y., Lopez-Gomez, I. & Zepeda-Núñez, L. Supporting software for 'Regional climate risk assessment from climate models using probabilistic machine learning'. *Zenodo* <https://doi.org/10.5281/zenodo.21268077> (2026).
47. Hersbach, H. et al. ERA5 hourly data on single levels from 1940 to present. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)* <https://doi.org/10.24381/cds.adbb2d47> (2023).
48. Liu, X., Gong, C. & Liu, Q. Flow straight and fast: learning to generate and transfer data with rectified flow. In *Proc. Eleventh International Conference on Learning Representations* <https://openreview.net/pdf/910c5efa5739a5d2bef83d432da87d3096712ebe.pdf> (2023).
49. Wood, A. W., Leung, L. R., Sridhar, V. & Lettenmaier, D. P. Hydrologic implications of dynamical and statistical approaches to downscaling climate model outputs. *Clim. Change* **62**, 189–216 (2004).
50. Wood, A. W., Maurer, E. P., Kumar, A. & Lettenmaier, D. P. Long-range experimental hydrologic forecasting for the eastern United States. *J. Geophys. Res. Atmos.* **107**, 4429 (2002).
51. Thrasher, B. et al. NASA global daily downscaled projections, CMIP6. *Sci. Data* **9**, 262 (2022).
52. Hayhoe, K., Scott-Fleming, I., Stoner, A. & Wuebbles, D. J. STAR-ESDM: a generalizable approach to generating high-resolution climate projections through signal decomposition. *Earth's Future* **12**, e2023EF004107 (2024).
53. Ullrich, P. A. *Validation of LOCA2 and STAR-ESDM Statistically Downscaled Products* Technical report https://eesm.science.energy.gov/sites/default/files/2024-12/ValidationDownscaledProducts%20v2_0.pdf (Lawrence Livermore National Laboratory, 2023).

Acknowledgements

We thank L. Li and S. Hoyer for productive discussions, and D. Worrall and J. Platt for feedback on the article. For the LENS2 dataset, we acknowledge the CESM2 Large Ensemble Community Project and the supercomputing resources provided by the IBS Center for Climate Physics in South Korea. ERA5 data⁴⁷ were downloaded from the Copernicus Climate Change Service. The results contain modified Copernicus Climate Change Service information. Neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains. We thank T. Russell for managing data acquisition and other internal business processes.

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L.Z.-N., F.S., Z.Y.W. and I.L.-G. conceptualized the work. F.S. and L.Z.-N. managed the project. R.C., Z.Y.W. and I.L.-G. curated the

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Funding

This work was supported by Google LLC.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s42256-026-01308-7>.

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Peer review information *Nature Machine Intelligence* thanks Matthias Karlbauer, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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